Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 40: As Few Edges As Possible

Fundamental Properties of Trees

Chartrand & Zhang, Section 13.1, pp. 491–497

Name: _____ Date: _____

Where we are

Lecture 39 ended with a bound: a connected graph of order n needs at least $n-1$ edges. Graphs that hit that bound exactly are called **trees**, and they turn out to be the most useful graphs there are — file systems, parse trees, decision trees, search trees.
Welcome back. One week of material remains.

By the end of today you should be able to

1. State the definition of a tree and recognise one.
2. Use the fact that a tree of order n has size $n-1$.
3. Apply the leaf-count formula.

1. Definitions

Definitions 13.1–13.3

A **tree** is a connected graph with no cycles.
A **forest** is a graph with no cycles — so a connected forest is a tree, and every component of a forest is a tree.
A **spanning tree** of a connected graph G is a spanning subgraph that is a tree: it uses every vertex of G and just enough edges to stay connected.

Which are trees? $V = \{1, 2, 3, 4\}$ in each case.

Edge set	Tree?	Why
$\{12, 23, 34\}$	Yes	Connected, no cycle. It is P_4.
$\{12, 13, 14\}$	Yes	Connected, no cycle. A star.
$\{12, 23, 31\}$	No	Has a cycle, and vertex 4 is isolated.
$\{12, 34\}$	No	No cycle, but disconnected — a forest.

2. Two leaves, always

Theorem 13.5

Every nontrivial tree contains at least two **leaves** (vertices of degree 1).

Proof. Let $P = (v_1, \ldots, v_k)$ be a longest path in the tree T. We claim $\deg v_1 = 1$.

Certainly $\deg v_1 \geq 1$, since v_1 is adjacent to v_2. Could v_1 have another neighbour?

- A neighbour v_0 *off* the path would give the longer path $(v_0, v_1, \ldots, v_k)$ — contradicting that P is longest.
- A neighbour v_t *on* the path with $t \geq 3$ would give the cycle $(v_1, v_2, \ldots, v_t, v_1)$ — impossible in a tree.

So $\deg v_1 = 1$, and the same argument gives $\deg v_k = 1$. ■

Why a longest path

Choosing an extremal object — longest, smallest, first — and showing it cannot be improved is the same move as Lecture 39's proof that a walk contains a path, and as the minimal counterexample arguments from Lecture 30.
When stuck on a graph proof, take something longest or smallest.

3. A tree has exactly $n-1$ edges

Theorem 13.6

Every tree of order n has size $n-1$.

Proof by induction on n.

Base step. The only tree of order 1 is K_1, with size $0 = 1-1$. ✓

Inductive step. Assume every tree of order k has $k-1$ edges, and let T be a tree of order $k+1$. Since T is nontrivial it has a leaf v, by Theorem 13.5. Then $T-v$ is a tree of order k, so by the hypothesis it has $k-1$ edges. Restoring v restores exactly one edge, so T has k edges. ■

Theorem 13.5 is load-bearing

The induction needs a leaf to delete. Without Theorem 13.5 there might be no leaf to remove, and the step would not run.
Notice what is happening: Week 11's induction, applied to a graph, using a result proved by the longest-path trick. The course's techniques are composing.

4. Any two of three

Theorem 13.10

If a graph G of order n and size m has **any two** of
(1) G is connected, (2) G has no cycles, (3) $m = n-1$,
then G is a tree — and therefore has the third as well.

One is not enough. On $V = \{1,2,3,4\}$ with $E = \{12, 23, 31\}$: the size is $3 = n-1$ ✓, but the graph is neither connected nor acyclic — a triangle plus an isolated vertex.

Theorem 13.11

G is a tree **if and only if** G has a *unique* u–v path for every two vertices u and v.

Two distinct u–v paths would together contain a cycle, which a tree cannot have.

5. Counting leaves

Theorem 13.12

Let T be a tree with n_i vertices of degree i. Then the number of leaves is

$$n_1 = 2 + n_3 + 2n_4 + 3n_5 + \cdots + (k-2)n_k.$$

Degree-2 vertices contribute nothing — they just subdivide an edge. Every vertex of degree 3 or more forces extra leaves.

Example. A tree has four vertices of degree 3 and two of degree 5, and no other vertices of degree ≥ 3.

How many leaves?

$$n_1 = 2 + 4 + 3(2) = \underline{12}$$

Order: $12 + 6 = \underline{18}$. Size: $\underline{17}$.

Check with the degree sum: $12(1) + 4(3) + 2(5) = 34 = 2(17)$. ✓

Exit ticket

1. A tree has order 23. What is its size?

2. A tree has six vertices of degree 3 and no other vertices of degree ≥ 3. How many leaves?

Before Wednesday

Read Section 13.2, "Rooted and Spanning Trees," pp. 498–505.

Self-check (not collected). Section 13.1, Exercises 1, 9, 11, and 13 — odd-numbered, answers in the back.

Coming up

Homework 14 is due Friday, December 4, 11:59 p.m. — the last assignment.
Friday is a cumulative review. Bring questions.
Final exam: December 7–11. Check Canvas for the time.

Questions I still have